IMPLEMENTATION PROOF · SOLO AI BUILD

# Field-to-GL: An AI-Built Mobile Labor Capture and Intercompany Billing Pipeline

Replaced a paper-timesheet and Excel-transcription process at RJK with a mobile web app (built with Replit Agent), an AI review layer (Shortcut.AI Skill), and a Power Query/Pivot ETL that produces a Yardi-ready intercompany labor payable file across 70+ property entities and 50+ active job-cost projects.

| ROLE | USERS | SCOPE | STACK |
|---|---|---|---|
| End-to-end design & build; rollout lead; monthly pipeline operator | 20+ field staff; 3 property managers; 3 construction PMs | 70+ property entities; 50+ job-cost projects; 12k+ entries | Replit Agent · Shortcut.AI Skill · Power Query/Pivot · Yardi |

## BEFORE › AFTER

### BEFORE — 7 STEPS

1. **Paper / ad-hoc digital timesheets**
   Field staff logged hours by hand or in scattered templates.
2. **Monday submission lag**
   Hours arrived a week late, detached from context.
3. **Jr. accounting re-keying**
   20–30 hrs/mo transcribing into Excel; notes lost.
4. **PM separate payroll pass**
   Property managers re-transcribed ~2 hrs/wk each.
5. **Monthly template consolidation**
   Manual roll-up across 70+ entities, 50+ projects.
6. **Rate & allocation math by hand**
   PMA escalators and charge types applied per-property.
7. **Sr. accounting review & Yardi entry**
   ~8 hrs senior effort to close monthly billing.

MONTHLY COST **~30–38 hrs**

### AFTER — 4 STEPS

1. **Mobile capture at point-of-work**
   Property, work type, detail, hours, notes — structured on submit.
2. **Weekly PM approval**
   Same-week review; approved entries feed payroll and billing from one source.
3. **AI review + deterministic ETL**
   Shortcut.AI Skill flags inconsistencies; Power Query maps rates & allocations.
4. **Yardi-ready CSV import**
   Intercompany payable file with entity routing & GL accounts, one-shot import.

MONTHLY COST **~50–65 min**

NET DELTA **~97% monthly time reduction · 3 fewer process steps · single source for payroll + billing**

**~$103K**
ESTIMATED ANNUAL OVERHEAD SAVINGS

**20+**
FIELD STAFF USERS

**50+**
JOB-COST PROJECTS

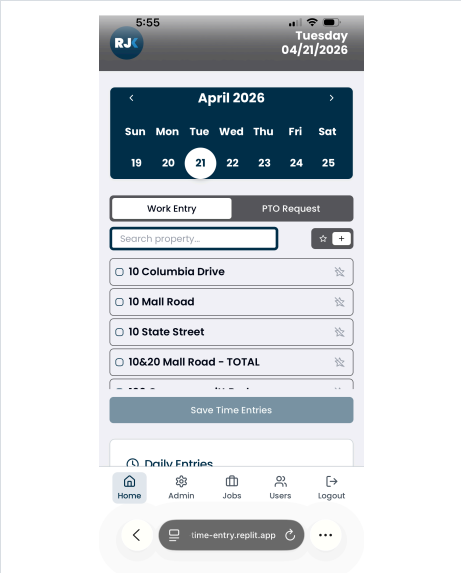
**70+**
PROPERTY ENTITIES

**12k+**
ENTRIES PROCESSED

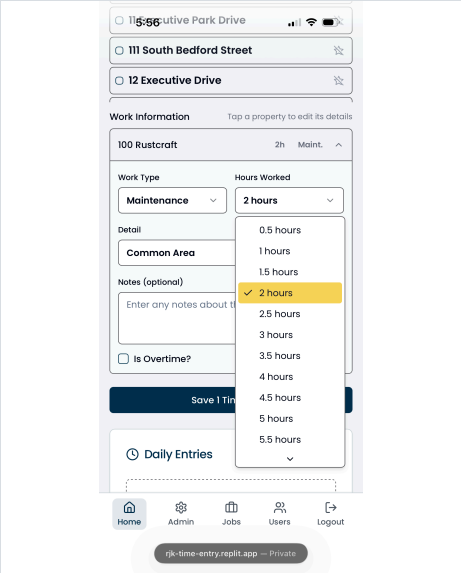
## WHAT I BUILT

- **Mobile web app** for labor time capture, engineered end-to-end with Replit Agent using deliberate prompt & context engineering. Structured inputs: property, work type (Maintenance / Construction), detail category (Common Area, Tenant Specific, Vacant Space, or active Project), hours, and free-text notes.
- **Approval layer** for property and construction project managers to review weekly entries and submit payroll-ready reports.
- **AI review layer** — a custom Shortcut.AI Skill that autonomously flags inconsistent entries using keyword rules and semantic mismatch against entry metadata.
- **ETL pipeline** in Power Query / Power Pivot for validation, rate mapping (by role and PMA escalator), charge-type allocation, and last-mile Excel formulas to produce a Yardi-importable CSV.
- **Controls**: required fields at capture, weekly PM approvals, monthly AI-assisted review, hour-total reconciliation, and a blended-rate sanity check on total billings.

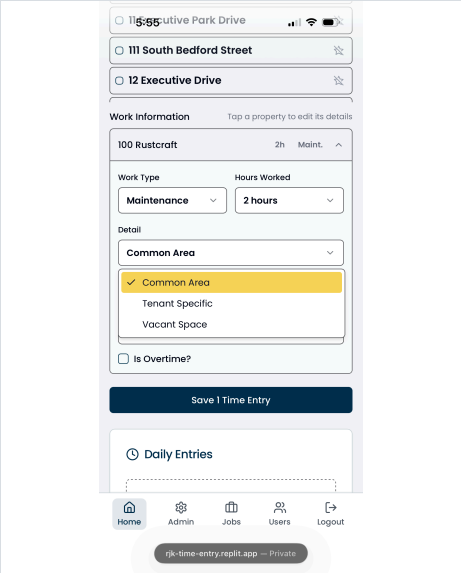
## CAPTURE — MOBILE-FIRST FIELD ENTRY




**STEP 1** Date auto-set, property selector.



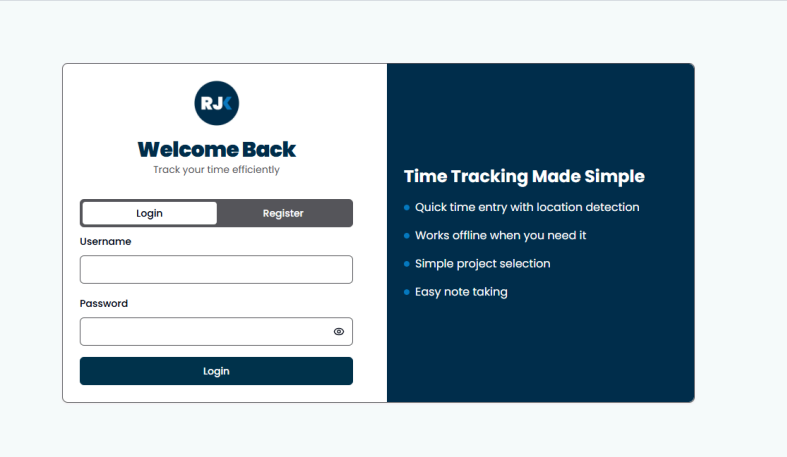

**STEP 2** Work type & detail category.




**STEP 3** Hours, notes, submit.

Synthetic sample data. All three flows submit into a single structured record preserved through the GL package.

## REVIEW — DESKTOP ENTRY POINT




**WELCOME** Desktop entry point — same authenticated user / PM flow as mobile. PMs approve weekly entries here; the same approved-entry set feeds both weekly payroll submissions and the monthly intercompany billing pipeline.

## TRANSFORM — DETERMINISTIC ETL

### INPUTS

Date (auto), property, work type, detail category (or active project for construction), hours, billable flag, PTO/holiday, and notes. Project managers maintain the active-project list per property directly in-app.

### PROCESSING LOGIC

Power Query normalizes the app export and joins a rate table (per role; adjustable for role changes and PMA inflation escalators). Power Pivot aggregates by property × charge type (Common Area, Tenant Specific, Vacant Space, Project) and routes to the correct JV entity. Last-mile Excel formulas produce a Yardi import layout: payee, invoice date/#, post month, entity, amount, GL / A/P accounts, notes.

### OUTPUT

A weekly payroll-submission report reviewed by property and construction PMs; and a monthly Yardi-ready intercompany payable CSV from the operator entity to each JV property entity, with job-cost project and cost-code detail preserved on construction entries.

| I | # OF PAYABLE SCREENS | PAYEE | INVOICE DATE | POST MONTH | INVOICE # | ENTITY | AMOUNT | ACCOUNT | A/P ACCOUNT | NOTES |
|---|---|---|---|---|---|---|---|---|---|---|
| I | 1 | llcrjkco | 2/28/2026 | 28-Feb | KFPARENT.L_022826 | 10skylin | 4320 | 51003200 | 20001000 | 02/2026 - RJK Labor |
| I | 1 | llcrjkco | 2/28/2026 | 28-Feb | KFPARENT.L_022826 | 10skylin_ | 1755 | 51003200 | 20001000 | 02/2026 - RJK Labor |
| I | 1 | llcrjkco | 2/28/2026 | 28-Feb | KFPARENT.L_022826 | 11orchar | 180 | 51003200 | 20001000 | 02/2026 - RJK Labor |
| I | 2 | llcrjkco | 2/28/2026 | 28-Feb | KFPARENT.IL_022826 | 12park | 1579.5 | 51003200 | 20001000 | 02/2026 - RJK Labor |
| I | 2 | llcrjkco | 2/28/2026 | 28-Feb | KFPARENT.IL_022826 | 12resear | 632 | 51003200 | 20001000 | 02/2026 - RJK Labor |
| I | 2 | llcrjkco | 2/28/2026 | 28-Feb | KFPARENT.IL_022826 | 12rivers | 5780 | 51003200 | 20001000 | 02/2026 - RJK Labor |
| I | 2 | llcrjkco | 2/28/2026 | 28-Feb | KFPARENT.IL_022826 | 135reseai | 757.5 | 51003200 | 20001000 | 02/2026 - RJK Labor |
| I | 2 | llcrjkco | 2/28/2026 | 28-Feb | KFPARENT.IL_022826 | 14center | 2282.5 | 51003200 | 20001000 | 02/2026 - RJK Labor |
| I | 2 | llcrjkco | 2/28/2026 | 28-Feb | KFPARENT.IL_022826 | 14pionee | 1642.5 | 51003200 | 20001000 | 02/2026 - RJK Labor |
| I | 2 | llcrjkco | 2/28/2026 | 28-Feb | KFPARENT.IL_022826 | 140harbo | 1597.5 | 51003200 | 20001000 | 02/2026 - RJK Labor |
| I | 2 | llcrjkco | 2/28/2026 | 28-Feb | KFPARENT.IL_022826 | 15corpor | 2385 | 51003200 | 20001000 | 02/2026 - RJK Labor |
| I | 2 | llcrjkco | 2/28/2026 | 28-Feb | KFPARENT.IL_022826 | 150waterf | 2902.5 | 51003200 | 20001000 | 02/2026 - RJK Labor |
| I | 3 | llcrjkco | 2/28/2026 | 28-Feb | 16corpor_022826 | 16corpor | 2666.25 | 51003200 | 20001000 | 02/2026 - RJK Labor |
| I | 4 | llcrjkco | 2/28/2026 | 28-Feb | 17orchar_022826 | 17orchar | 360 | 51003200 | 20001000 | 02/2026 - RJK Labor |
| I | 5 | llcrjkco | 2/28/2026 | 28-Feb | 175memori_022826 | 175memo | 2227.5 | 51003200 | 20001000 | 02/2026 - RJK Labor |
| I | 6 | llcrjkco | 2/28/2026 | 28-Feb | 18maple_022826 | 18maple | 2902.5 | 51003200 | 20001000 | 02/2026 - RJK Labor |
| I | 6 | llcrjkco | 2/28/2026 | 28-Feb | 18maple_022826 | 18maple | 337.5 | 59008000 | 20001000 | 02/2026 - RJK Labor |
| I | 7 | llcrjkco | 2/28/2026 | 28-Feb | 20summit_022826 | 20summit | 630 | 51003200 | 20001000 | 02/2026 - RJK Labor |
| I | 8 | llcrjkco | 2/28/2026 | 28-Feb | 22atlant_022826 | 22atlant | 675 | 51003200 | 20001000 | 02/2026 - RJK Labor |
| I | 9 | llcrjkco | 2/28/2026 | 28-Feb | 22maple_022826 | 22maple | 2430 | 51003200 | 20001000 | 02/2026 - RJK Labor |
| I | 9 | llcrjkco | 2/28/2026 | 28-Feb | 22maple_022826 | 22maple | 180 | 59008000 | 20001000 | 02/2026 - RJK Labor |
| I | 10 | llcrjkco | 2/28/2026 | 28-Feb | 220harbor_022826 | 220harbc | 1215 | 51003200 | 20001000 | 02/2026 - RJK Labor |
| I | 11 | llcrjkco | 2/28/2026 | 28-Feb | 25atlant_022826 | 25atlant | 3387.5 | 51003200 | 20001000 | 02/2026 - RJK Labor |

**OUTPUT** Yardi-ready intercompany payable CSV — the final artifact the ETL produces. Each row is a property-entity payable from the operator entity, with GL account, A/P account, and notes populated for a one-shot monthly import.

## AI IN THE BUILD

AI is used as leverage at three specific seams. Durable transformation logic sits in deterministic Power Query / Power Pivot; AI accelerates build, review, and codification.

**REPLIT AGENT**

### Spec-to-app in hours

Prompt- and context-engineered the full-stack web app: schema, auth, routing, entry form, PM approval surface, admin. Solo build, no engineering team.

**SHORTCUT.AI SKILL**

### Autonomous review layer

Custom Agent Skill developed with examples of inconsistent entries. Flags keyword and semantic mismatches between notes and metadata before senior review.

**POWER QUERY / PIVOT**

### AI-assisted ETL design

Rate mapping, PMA escalator math, charge-type routing, and blended-rate sanity check logic drafted with AI, then hardened into deterministic M-code and DAX.

**Personal AI capability on display:** spec-writing for agentic builders, custom skill training, prompt-driven ETL design, and the judgment to keep the *posting* path human-gated — AI for leverage, deterministic code for audit integrity.

## BUSINESS IMPACT — DETAIL

**~$103K**

ESTIMATED ANNUAL OVERHEAD SAVINGS

- Sr. accounting monthly close effort for labor billing: ~8 hrs → ~30–45 min.
- Jr. accounting transcription eliminated (~20–30 hrs/month).
- Property manager timesheet transcription: ~2 hrs/week → ~20 min/week review.
- Source: *OverHead Savings Calculation.xlsx* (internal, headcount × loaded hourly rate × hours saved).

| 20+ | 70+ | 50+ | 12k+ |
|---|---|---|---|
| FIELD STAFF USERS | PROPERTY ENTITIES BILLED | ACTIVE JOB-COST PROJECTS | TIME ENTRIES PROCESSED |

### QUALITATIVE IMPACT

- **Cleaner billing:** standardized charge-type routing and blended-rate sanity checks replaced ad-hoc spreadsheet work.
- **Faster close support:** end-to-end monthly billing collapses to under an hour of senior review.
- **Richer entry data:** per-entry notes from the field are preserved into the GL package — previously lost in manual re-keying.
- **Better accountability:** weekly PM approval and AI-assisted inconsistency flagging sit in front of monthly billing.

### ADOPTION & CONTINUITY

- Rolled out to the field, property, and construction teams; in continuous monthly use for 10+ months with no reported downtime.
- Operator ran the monthly pipeline personally; process remained in place at RJK after departure.
- Reused as both the payroll submission source and the monthly intercompany billing source from a single dataset.

## TRANSFERABLE RELEVANCE

- **End-to-end implementation judgment:** field capture, review, ETL, and ERP integration designed as one system — not a point tool bolted on.
- **Business-process understanding:** deliberate design of inputs, approvals, reconciliation, and audit-friendly outputs for finance, property, and construction stakeholders.
- **AI / tooling fluency:** agentic build tools and custom skills used where they pay; deterministic code where audit integrity matters.
- **Finance-ops relevance:** eliminates manual re-keying, tightens close, and feeds ERP postings with structured, audit-defensible data.

## LIMITATIONS & CONTROLS ACKNOWLEDGED

- Monthly human review is required before posting; the AI review layer flags, it does not auto-correct.
- Rate tables and PMA escalators are maintained manually; changes flow through a single rate source.
- Yardi posting step is a controlled import — intentionally human-gated, not automated end-to-end.

# APPENDIX — SUPPORTING EVIDENCE

**Overhead Savings Calculator**

**Assumptions**

Working Hours per Year 2,080

| Role | Headcount | Annual Salary | Load Factor | Loaded Salary | Hours Saved / Month | Loaded Hourly Rate | Monthly Savings | Annual Savings |
|---|---|---|---|---|---|---|---|---|
| Jr. Accounting Staff | 1 | $ 70,000.00 | 1.22 | $ 85,400.00 | 30 | $ 41.06 | $ 1,231.73 | $ 14,780.77 |
| Sr. Accounting Staff | 1 | $ 185,000.00 | 1.22 | $ 225,700.00 | 8 | $ 108.51 | $ 868.08 | $ 10,416.92 |
| Property Manager | 3 | $ 140,000.00 | 1.22 | $ 170,800.00 | 4 | $ 82.12 | $ 985.38 | $ 11,824.62 |
| Construction Project Manager | 3 | $ 185,000.00 | 1.22 | $ 225,700.00 | 4 | $ 108.51 | $ 1,302.12 | $ 15,625.38 |
| Construction/Maintenance Field Staff | 20 | $ 90,000.00 | 1.22 | $ 109,800.00 | 4 | $ 52.79 | $ 4,223.08 | $ 50,676.92 |
| **Total Savings** | | | | | **50** | | **$ 8,610.38** | **$ 103,324.62** |

**IMPACT MATH** Overhead Savings Calculator — headcount × loaded hourly rate × hours saved per month, rolled to annual (~$103K). Source workbook: internal; synthetic loading factors shown.

| A | B | C | D | E | F | G | H | I |
|---|---|---|---|---|---|---|---|---|
| | bill_type | All | | | | | | |
| | | | | | | | | |
| | | Column Labels | | | | | | |
| | | COMMON_AREA | | VACANT_SPACE | | Total Sum of billing_amt | Total Sum | |
| | Row Labels | Sum of billing_amt | Sum of adjusted_hours | Sum of billing_ar | Sum of adjusted_hours | | | |
| | 10 Skyline Way (Apex) | $1,012.50 | 28.5 | | | $1,012.50 | 28.5 | |
| | 10 Skyline Way (Pinnacle) | $270.00 | 8.5 | | | $270.00 | 8.5 | |
| | 18&20 Maple Drive | $11,058.75 | 162 | | | $11,058.75 | 162 | |
| | 11 Orchard Drive | $11,913.75 | 176.5 | | | $11,913.75 | 176.5 | |
| | 12 Park Avenue | $2,565.00 | 37 | | | $2,565.00 | 37 | |
| | 12 Research Drive | $10,957.50 | 160 | | | $10,957.50 | 160 | |
| | 12 Riverside Road | $405.00 | 6 | | | $405.00 | 6 | |
| | 135 Research Drive | $180.00 | 2 | | | $180.00 | 2 | |
| | 14 Center Street | $2,385.00 | 34 | | | $2,385.00 | 34 | |
| | 14 Pioneer Drive | $270.00 | 11 | | | $270.00 | 11 | |
| | 140 Harbor View Drive | $1,822.50 | 28 | | | $1,822.50 | 28 | |
| | 15 Corporate Park Drive | $1,755.00 | 26 | | | $1,755.00 | 26 | |
| | 150 Waterfront | $7,965.00 | 118 | | | $7,965.00 | 118 | |
| | 16 Corporate Drive | | 6 | | | | 6 | |
| | 17_21_Orchard | $67.50 | 1 | | | $67.50 | 1 | |

**REVIEW SURFACE** Pivot view used during monthly review — per-property billing and hours by charge type, used as the blended-rate sanity check before Yardi import (synthetic sample data).